LAB EXPERIMENT

18. Implement image cropping, copying and pasting to select a region of interest (ROI) from the source image using Open CV.

Code:

```
import cv2

# Read the image
img = cv2.imread(r"C:\Users\Admin\Downloads\images.jpg")

if img is None:
    print("Error: Image not found!")
else:
    # Crop the Region of Interest (ROI)
    roi = img[50:200, 50:200]

    # Copy the original image
    copied = img.copy()

    # Get ROI dimensions
    h, w = roi.shape[:2]

    # Paste ROI at a valid location
    x, y = 0, 0
    copied[y:y+h, x:x+w] = roi

    # Display images
    cv2.imshow("Original Image", img)
    cv2.imshow("Cropped ROI", roi)
    cv2.imshow("Copied and Pasted Image", copied)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```


Original Image


Cropped ROI


Copied and Pasted Image